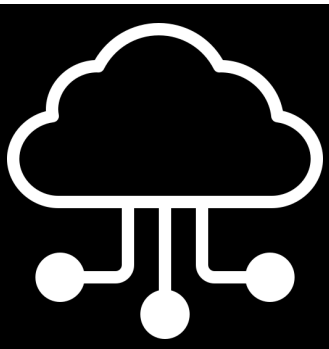


Introduction to Cloud Computing

Lecture 01

Practical Cloud Computing

Sina Keshvadi
Thompson Rivers University

Course objectives

- Apply Amazon Web Services (AWS) architectural principles and best practices to make architectural decisions.
- Use appropriate AWS services and features to make infrastructure scalable, reliable, and highly available.
- Choose AWS managed services to enable greater flexibility and resiliency in an infrastructure.
- Increase performance and reduce cost of a cloud infrastructure built on AWS.
- Use AWS services and features to secure user, application, and data access.
- Apply best practices from the AWS Well-Architected Framework to improve architectures that use AWS solutions.

Throughout the course, you will learn about the architectural layers, design principles, and Amazon Web Services (AWS) services used in three general architecture patterns as follows:

- Three-tier architectures
- Serverless architectures
- Streaming architectures

Alignment to AWS Certification

- The following knowledge and skills are validated by the AWS Certified Solutions Architect exam:
- Ability to design solutions based on the AWS Well-Architected Framework
- Ability to perform the following tasks:
  - Design solutions that incorporate AWS services to meet current business requirements and future projected needs.
  - Design architectures that are secure, resilient, high-performing, and cost-optimized.
  - Review existing solutions and determine improvements.



3

This course is not an exam preparation course for AWS Certification. However, the lectures and labs are designed to align with the knowledge and skills described in the AWS Certified Solutions Architect – Associate exam (SAA-C03) guide:
https://docs.aws.amazon.com/aws-certification/latest/solutions-architect-associate-03/solutions-architect-associate-03.html

Course outline by lectures

1. Introducing Cloud Architecting
2. Securing Access
3. Adding a Storage Layer with Amazon S3
4. Adding a Compute Layer Using Amazon EC2
5. Adding a Database Layer
6. Creating a Networking Environment
7. Connecting Networks
8. Securing User, Application, and Data Access
9. Implementing Monitoring, Elasticity, and High Availability
10. Automating Your Architecture
11. Caching Content
12. Building Decoupled Architectures
13. Building Serverless Architectures and Microservices
14. Data Engineering Patterns
15. Planning for Disaster
16. Serverless App Development Project

The evolving café architecture (Course Project)

Version	Business Reason for Update	Technical Requirements and Architecture Update
V1	Build a static website for a small business.	Host the website on Amazon S3.
V2	Update the website to support dynamic content and online ordering.	Deploy the web application and database on Amazon EC2.
V3	Reduce the effort to maintain the database and secure its data.	Separate web and database layers. Migrate the database to Amazon RDS on a private subnet.
V4	Enhance the security of the web application.	Use Amazon VPC features to configure and secure public and private subnets.
V5	Ensure that the website can handle an expected increase in traffic and remain highly available and resilient to failure.	Add a load balancer, implement auto scaling on the EC2 instances, and distribute compute and database instances across two Availability Zones.
V6	Automate deployments so that the café can consistently deploy, manage, and update café resources across Regions.	Build a version-controlled CloudFormation template to deploy the network and application layers. Deploy the CloudFormation stack to another Region.
V7	Add reporting capabilities while reducing the operational and maintenance burden, improving performance, and reducing costs.	Deploy Lambda functions that connect to the Amazon RDS database and generate a report based on a schedule.



5

Across the challenge labs in the course, you help Frank and Martha evolve their cloud architecture. You take it from a simple static website in the cloud to a secure, scalable, and resilient cloud-based application that helps them optimize costs. This table summarizes the updates you make in each challenge lab.

Section
The Cloud Architect and the Cloud Engineer

IT professional

Common skills and responsibilities include the following:

- Act as a generalist and might manage an application
- Often manage a production environment
- Highly technical
- Might have significant or limited experience in cloud technologies
- Might specialize in one area (such as security or storage)

Job titles include the following:

- IT administrator
- Systems administrator
- Network administrator

IT leader

Common skills and responsibilities include the following:

- Lead a team of IT professionals
- Responsible for day-to-day operations
- Manage a budget, stay informed about technologies, and choose new technologies
- Hands on during the early stages of a project, and then delegate the team to take over

Job titles include the following:

- IT manager
- IT director
- IT supervisor

IT leaders are managers. They typically lead a team of IT professionals and decide on the type of technology to use for a project. They might be significantly involved in the implementation details early in the project lifecycle. Then, they delegate the team to handle the details as the project gets closer to completion.

Developer

Common skills and responsibilities include the following:

- Write, test, and fix code
- Think about projects at the application level
- Work with APIs and SDKs
- Might use sample code
- Might specialize in one area (such as security or storage)

Job titles include the following:

- Software developer
- System architect
- Software development manager

Another common role in cloud computing is the developer role. Developers work with code. They work on the details—writing, testing, and fixing the code that makes an application work. Developers build and consume documentation and code repositories to accelerate projects. They work with APIs and SDKs.

DevOps engineer

Common skills and responsibilities include the following:

- Build out the infrastructure that applications run on, often in the cloud
- Follow the guidelines of the cloud architect
- Experiment to improve deployments

Job titles include the following:

- DevOps engineer
- Build engineer
- Reliability engineer

Development Operations (DevOps) engineers build out the infrastructure that applications run on. They often create or improve on the code that installs and configures servers and application deployments. DevOps staff often experiment with the infrastructure and code. Sometimes, they learn by doing. They create repeatable deployment solutions, and they work to apply engineering skills to the business needs of operations teams.

Cloud Architect is like the **architect** who designs the blueprint for the house.
Cloud Engineer is like the **construction crew** that builds the house based on the blueprint.

Scenario: The Growing E-Commerce Platform

The Company: A mid-sized online retail company called "StyleUp" has been running their e-commerce website on a single, powerful server located in their office building (an on-premises setup).

The Business Problem: StyleUp is launching a major marketing campaign for the upcoming holiday season. They anticipate a **10x increase in website traffic**. Their current server cannot handle this load; it will crash, leading to lost sales and unhappy customers. They need a new infrastructure solution that can handle the traffic spike, be highly available (never go down), and not break the bank. They've decided to move to Cloud.

The Task for the Cloud Architect: The CEO of StyleUp comes to you, the Cloud Architect, and says:

> *"We need to move our website to AWS before the holiday season in three months. It needs to be able to handle a massive, unpredictable amount of traffic without crashing. We also have a limited budget, so we can't just over-provision a bunch of massive servers. Finally, our customer data must be secure. Design a solution for us."*

The Cloud Architect

- Deconstruct the Requirements:
  - High Availability & Resilience
  - Scalability & Performance
  - Cost-Optimization
  - Security
- Design the Architecture
  - Network Foundation
  - Compute (Web Servers)
  - Database
- Data & Security

The Cloud Engineer

- Build the Network
- Set Up the Servers
- Migrate the Database
- Implement Security
- Configure Storage and CDN
- Automate Deployment
- Set Up Monitoring

As a cloud architect designing AWS architectures:

- I need to understand the practice of applying cloud best practices to a solution that uses cloud services and features to meet an organization's technical needs and business use cases.
- I need to evaluate an architecture with the AWS Well-Architected Framework.
- I need to apply best practices to building solutions on AWS.

This slide asks you to take the perspective of a cloud architect. Review these considerations as you progress through this module, remembering that the cloud architect should work backward from the business need to design the best architecture for a specific use case.

Key Responsibilities

Solutions architects have in-depth knowledge of the architectural principles and services and focus on designing, implementing, and managing technical solutions.

Key Responsibilities

- Cloud Strategy Development
  - Create the organization's cloud adoption roadmap and architecture vision.
- Solution Design
  - Design scalable, cost-effective, and secure cloud infrastructure.
- Technology Selection
  - Evaluate and select appropriate cloud providers and services.
- Migration Planning
  - Develop strategies to transition from on-premises to cloud environments.
- Security & Compliance
  - Ensure cloud implementations meet security standards and regulatory requirements.
- Cost Optimization
  - Balance performance needs with budget constraints.
- Mentorship
  - Guide development teams on cloud best practices.

Required Skills

A Solutions Architect should possess a diverse set of technical and soft skills.

- Deep technical knowledge of cloud platforms and services.
- Understanding of infrastructure-as-code.
- Network architecture and security expertise.
- Strong communication skills to translate between business and technical teams.
- Knowledge of DevOps practices and CI/CD pipelines.

Tools and Technologies

AWS Solutions Architects employ various tools and technologies to design effective cloud infrastructures.

These tools are combined based on specific project requirements and organizational preferences.

Core Design Tools

- **AWS Console**
  - Web-based management interface.
- **AWS CLI**
  - Command-line interface for AWS service management.
- **AWS CloudFormation**
  - Infrastructure as Code (IaC) service using template-based resource provisioning.
- **AWS CDK** (Cloud Development Kit)
  - Infrastructure as code using familiar programming languages.
- **AWS Diagrams/Architecture Center**
  - Reference architectures and best practices.

Infrastructure Planning and Management

- AWS Well-Architected Tool
  - Reviews architectures against AWS best practices.
- AWS Trusted Advisor
  - Real-time guidance for optimizing AWS environments.
- AWS Cost Explorer
  - Cost estimation and optimization tool.
- AWS Organizations
  - Account management and consolidated billing.
- AWS Control Tower
  - Set up and govern multi-account environments.

Monitoring and Security

- AWS CloudWatch
  - Monitoring and observability service.
- AWS CloudTrail
  - Governance, compliance, and audit capabilities.
- AWS Config
  - Resource inventory and configuration history tracking.
- AWS Security Hub
  - Security posture management.

Deployment and DevOps

- AWS CodePipeline/AWS CodeDeploy/AWS CodeBuild
  - Continuous integration/deployment.
- AWS Elastic Beanstalk
  - Platform as a Service (PaaS) for application deployment.

Third-Party Tools

- Terraform
  - Multi-cloud Infrastructure as Code (IaC) platform.
- Draw.io or Lucidchart
  - Architecture diagramming tools.
- Git repositories
  - Version control for infrastructure code.

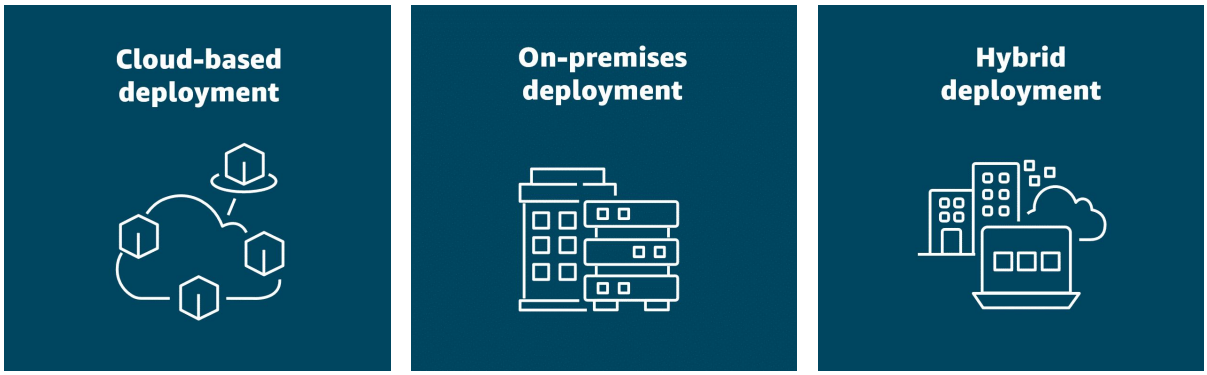
Introduction to the Cloud

Defining cloud computing



- **On-demand delivery**
  On-demand delivery means that customers can access computing resources, such as storage or compute power, within seconds and as needed. Users can scale their resource usage up or down based on current requirements without lengthy provisioning processes.
- **of IT resources**
  The of IT resources aspect highlights the wide array of information technology assets in the cloud-computing space. These resources include servers, storage solutions, databases, networking components, artificial intelligence and machine learning (AI/ML) tools, and more. Customers can use these resources to build, deploy, and manage applications and services through the cloud infrastructure.
- **over the internet**
  Over the internet signifies that cloud computing delivers IT resources through internet connectivity. This means that users access and use these resources through web-based services rather than maintaining local hardware or software. The internet acts as the conduit, which provides remote access to compute power, storage, and applications from anywhere in the world.
- **with pay-as-you-go-pricing**
  Flexible pricing is a fundamental economic aspect of cloud computing. Users pay only for the resources they actually consume, rather than committing to fixed, long-term contracts. This usage-based pricing model offers cost efficiency and financial flexibility.

Cloud deployment types



- **Cloud deployment types**
- You can deploy cloud resources in multiple ways: cloud, on-premises, and hybrid. Each type offers unique benefits and considerations, and exploring these options can help you make informed decisions about your cloud strategy.

- **Cloud**
- In a cloud-based deployment model, you have the flexibility to migrate your existing resources to the cloud, design and build new applications within the cloud environment, or use a combination of both.
- For instance, a company might migrate data resources to the cloud, then develop an application comprised of virtual servers, databases, and networking components entirely hosted in the cloud.

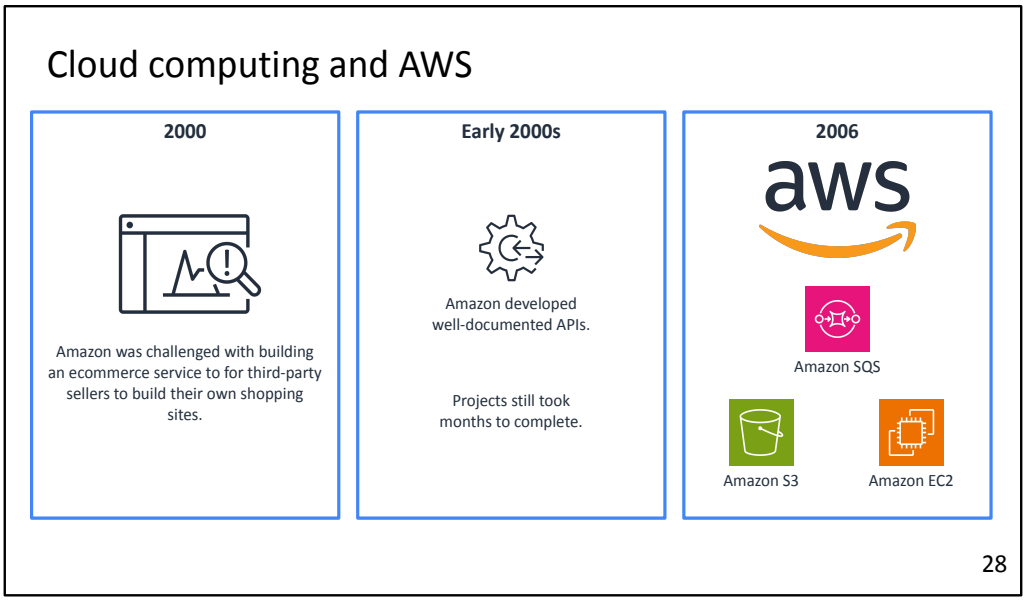
- **On-premises**
- Deploying resources on premises using virtualization and resource management tools does not provide many of the benefits of cloud computing. However, it is sometimes sought for its ability to provide dedicated resources and low latency.

- In most cases this deployment model is the same as legacy IT infrastructure while using application management and virtualization technologies to try increasing resource utilization.

- **Hybrid**
- In a hybrid deployment, cloud-based resources and on-premises infrastructure work together. This approach is ideal for situations where legacy applications must remain on premises due to maintenance preferences or regulatory requirements.

- For instance, a company might choose to retain certain regulated legacy applications on-premises while using cloud services for advanced data processing and analytics.

- Multi-cloud deployments can also be considered hybrid deployments.



To understand what cloud architecting is and why it's important, first consider an example of what software development is like in its absence. Around 2000, when Amazon was trying to solve a problem, they looked to cloud computing as a solution. Amazon was trying to build an ecommerce service for third-party sellers to build their own online shopping sites on top of the Amazon ecommerce engine.

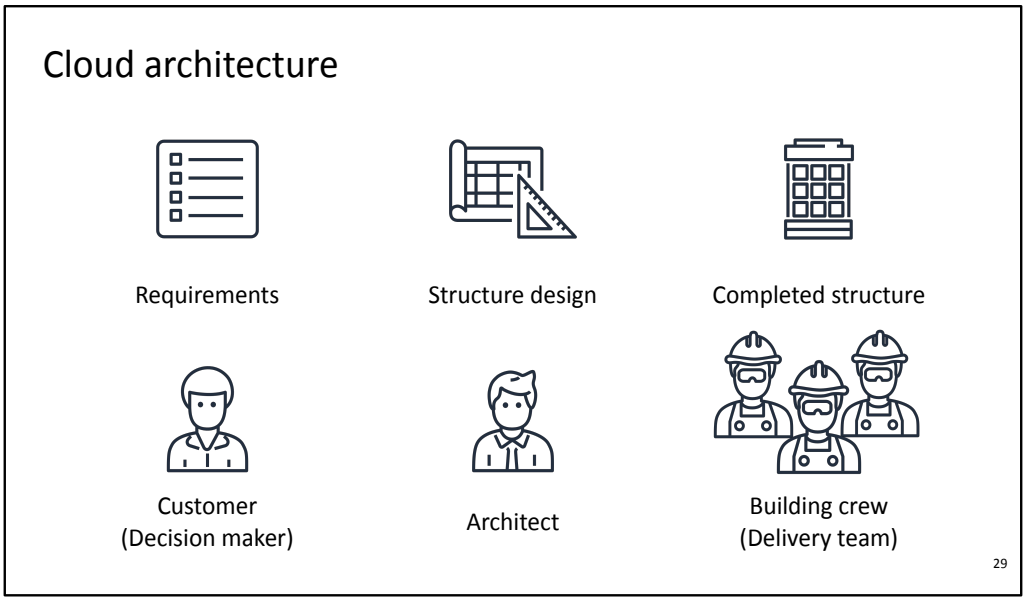
The company was struggling to make its new shopping website highly available and scalable. In the beginning, the Amazon ecommerce tools, applications, and architectures were built without proper planning. The various services were difficult to separate to make a centralized development platform. Development time was long, and projects were complicated.

The solution to this problem was to create a set of well-documented APIs to organize the development environment.

However, Amazon still struggled to build applications quickly. As the company grew and more software engineers were hired, the following complications happened:

- It took 3 months to build just the database, compute, and storage components when the entire project that was expected to take 3 months.
- Each team built their own resources without planning for scalability or reusability.

The solution was to build internal services to create highly available, scalable, and reliable architectures on top of the Amazon infrastructure. In 2006, Amazon started selling these services as Amazon Web Services (AWS). They started with Amazon Simple Queue Service (Amazon SQS), and then Amazon Simple Storage Service (Amazon S3) and Amazon Elastic Compute Cloud (Amazon EC2).

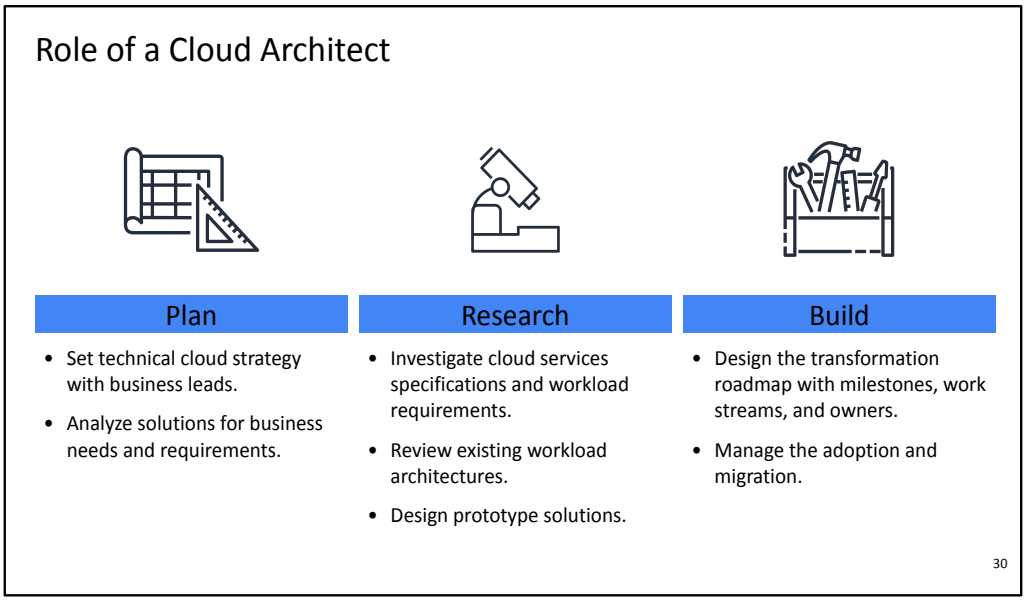


What is cloud architecture? Cloud architecture is the practice of applying cloud characteristics to a solution that uses cloud services and features to meet an organization's technical needs and business use cases.

Creating technology solutions is a lot like constructing a physical building. If the foundation is not solid, it can cause structural problems that undermine the integrity and function of the building. To create a solid building, the customer outlines the building needs and requirements. The architect creates a design and blueprints to meet the requirements. Then, the building crew turns the blueprints into a physical structure.

In cloud architecture, the customer or decision maker also outlines business goals and requirements. The cloud architect designs a solution that is like a blueprint for a building. Then, the delivery team works to implement the solution.

Having well-architected systems increases the likelihood that the technology deliverables will help meet business goals.



The role of a cloud architect is as follows:

- Engage with decision makers to identify the business goals and the capabilities that need improvement.
- Ensure alignment between the technology deliverables of a solution and the business goals.
- Work with the delivery teams that are implementing the solution to ensure that the technology features are appropriate.

Cloud architects are responsible for managing an organization's cloud computing architecture. They have in-depth knowledge of the architectural principles and services used to do the following:

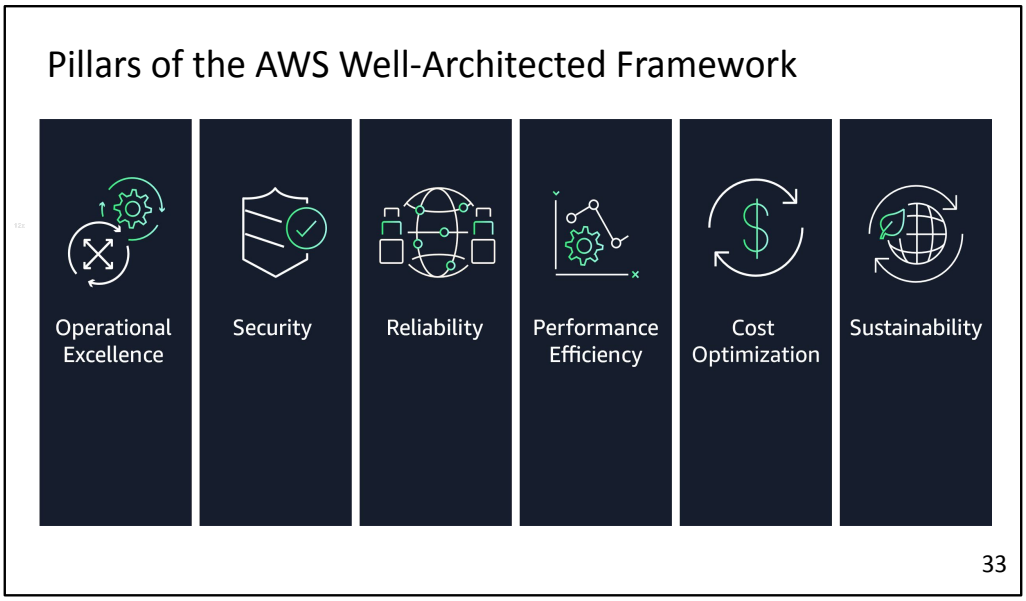
- Develop the technical cloud strategy based on business needs.
- Assist with cloud migration efforts.
- Review workload requirements.
- Provide guidance about how to address high-risk issues.

Cloud architects work to implement the AWS Well-Architected Framework.

Key takeaways: What is cloud architecting?

- Cloud architecture is the practice of applying cloud characteristics to a solution that uses cloud services and features to meet an organization's technical needs and business use cases.
- You can use AWS services to create highly available, scalable, and reliable architectures.
- Cloud architects are responsible for managing an organization's cloud computing architecture.

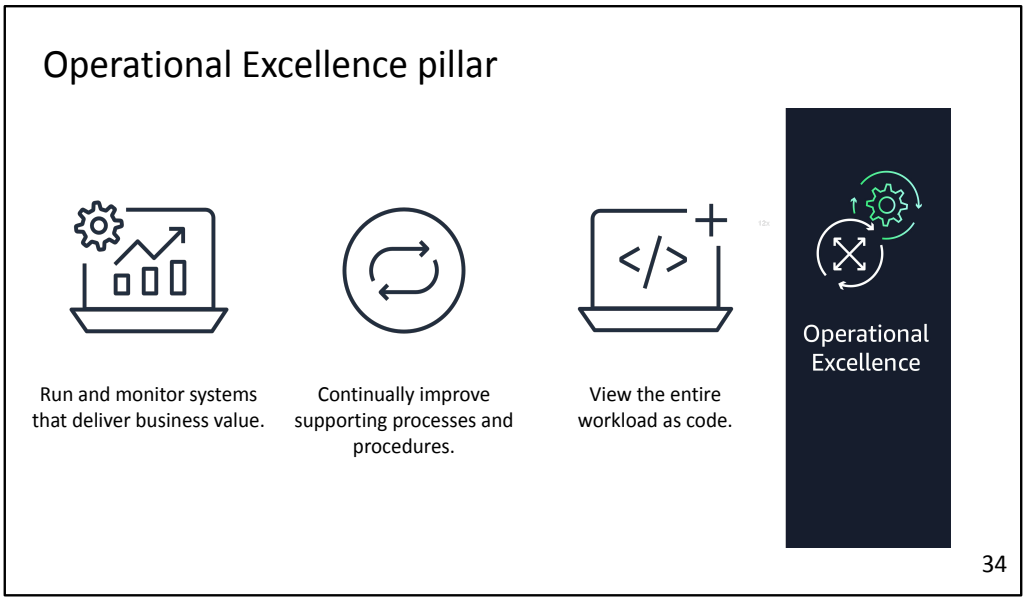
Section
AWS Well-Architected Framework



The AWS Well-Architected Framework is a guide that provides a consistent approach to evaluate cloud architectures. It also provides guidance to help implement designs. It documents a set of foundational questions and best practices you can use to understand if a specific architecture aligns well with cloud best practices. AWS developed this framework after reviewing thousands of customer architectures on AWS.

The AWS Well-Architected Framework is organized into six pillars: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, and Sustainability.

You revisit the pillars of the AWS Well-Architected Framework in each of the modules of this course. For more information, see the AWS Well-Architected Framework on the content resources page of your online course.

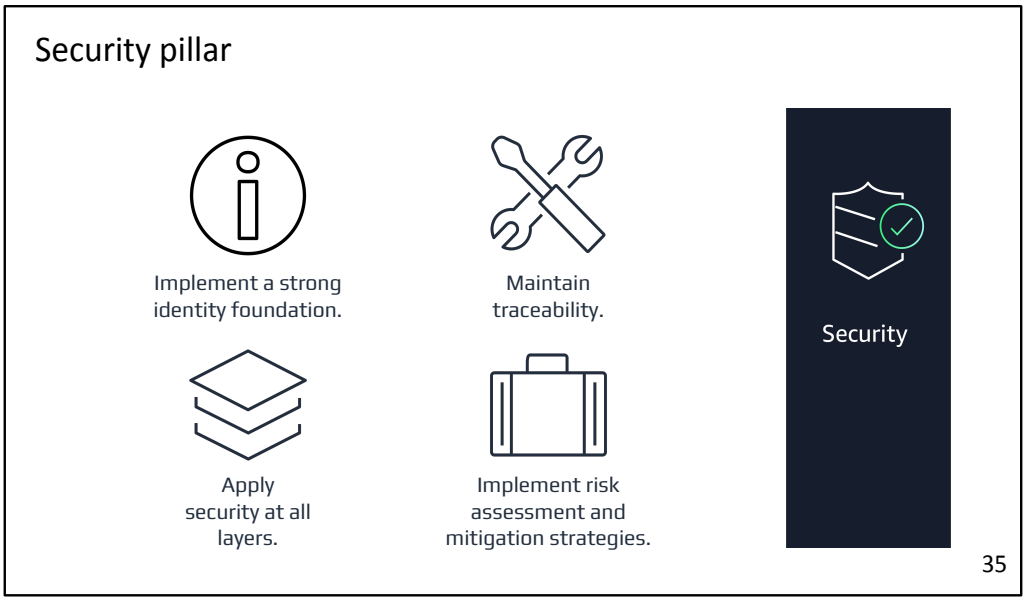


The Operational Excellence pillar addresses the ability to run systems and gain insight into their operations to deliver business value. It also addresses the ability to continuously improve supporting processes and procedures.

When you design a workload for operations, you must be aware of how it will be deployed, updated, and operated. Implement engineering practices that align with defect reductions and quick, safe fixes. Make observation possible with logging, instrumentation, and business and technical metrics so you can gain insight into what is happening inside your architecture.

In AWS, you can view your entire workload (applications, infrastructure, policies, governance, and operations) as code. You can define and update all part of your workload using code. This means you can apply the same engineering discipline that you use for application code to every element of your stack. It is recommended that you invest in implementing operations activities as code to maximize productivity, minimize error rates, and set up automated responses.

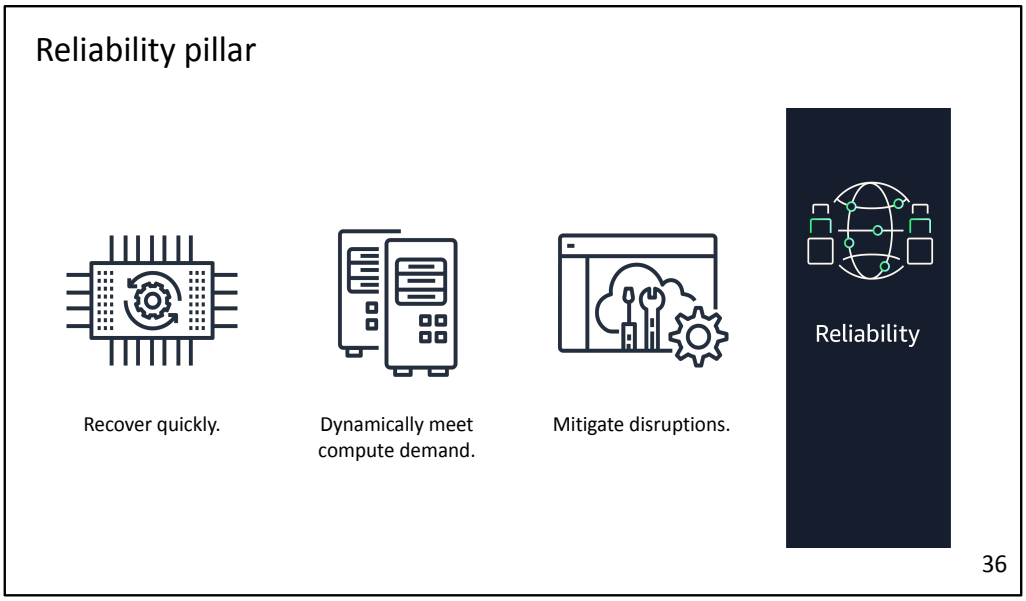
For more information, see the Operational Excellence pillar on the content resources page of your online course.



The Security pillar addresses the ability to protect information, systems, and assets while delivering business value through risk assessments and mitigation strategies.

Your architecture will present a much stronger security presence if you implement a strong identity foundation, use and maintain traceability, apply security at all layers, automate security best practices, and protect data in transit and at rest. Implementing these security principles helps you prepare for security events.

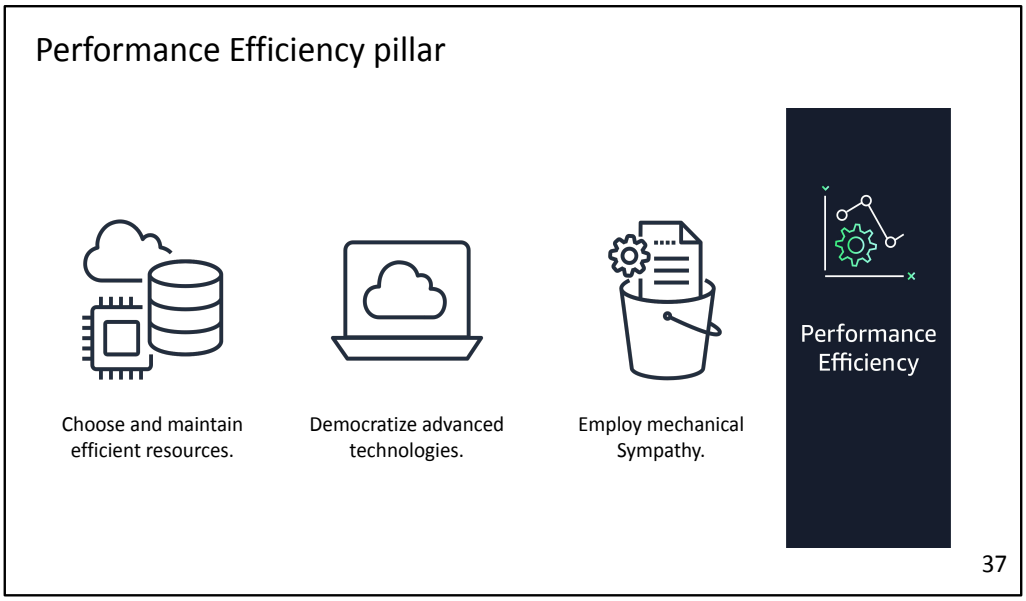
For more information, see the Security pillar on the content resources page of your online course.



The Reliability pillar addresses the ability of a system to recover from infrastructure or service disruptions and dynamically acquire computing resources to meet demand. It also addresses the ability of a system to mitigate disruptions, such as misconfigurations or transient network issues.

It can be difficult to ensure reliability in a traditional environment. Issues arise from single points of failure, lack of automation, and lack of elasticity. By applying the best practices outlined in the Reliability pillar, you can prevent many of these issues. The Reliability pillar helps you and your customers have a properly designed architecture with respect to high availability, fault tolerance, and overall redundancy.

For more information, see the Reliability pillar on the content resources page of your online course.

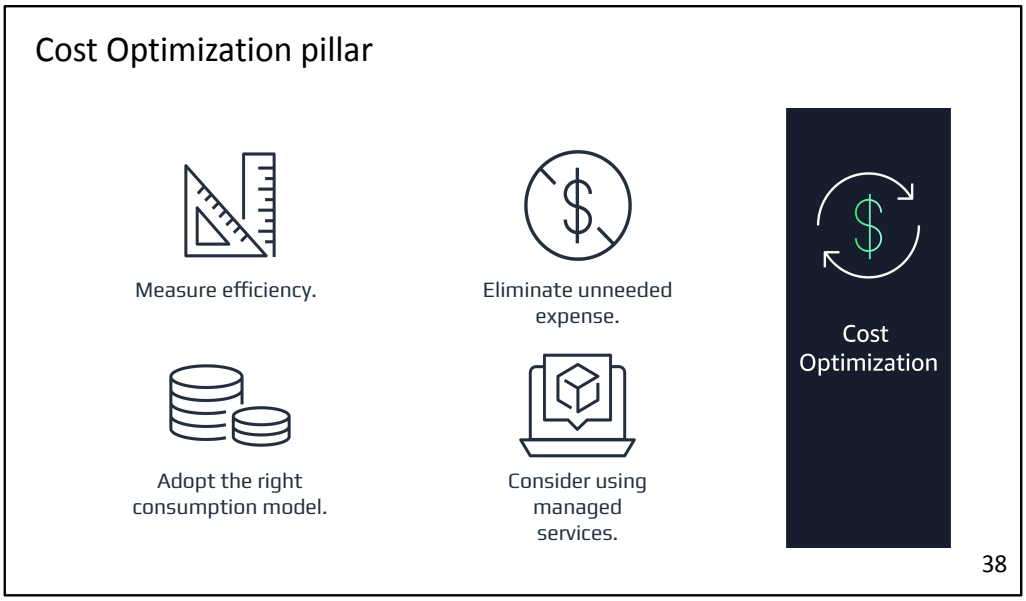


When you consider performance, you want to maximize your performance by using computation resources efficiently. You also want to maintain that efficiency as the demand changes.

It is also important to democratize advanced technologies. In situations where technology is difficult to implement yourself, consider using a vendor. By implementing the technology for you, the vendor handles the complexity and the knowledge, freeing your team to focus on more value-added work.

Mechanical sympathy is when you use a tool or system with an understanding of how it operates best. Use the technology approach that aligns best to what you are trying to achieve. For example, consider data access patterns when you select database or storage approaches.

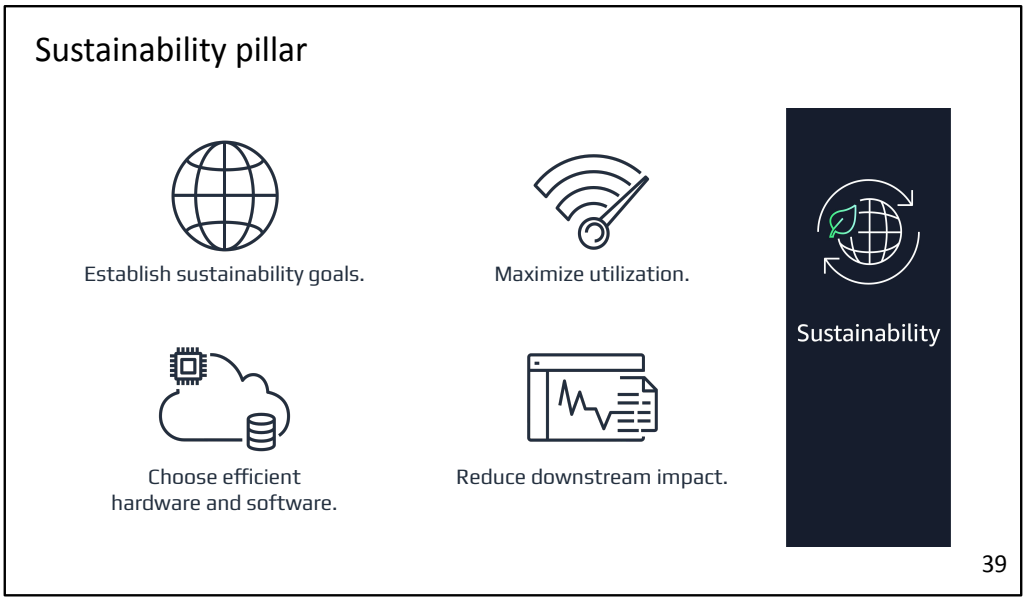
For more information, see the Performance Efficiency pillar on the content resources page of your online course.



Cost optimization is an ongoing requirement of any good architectural design. The process is iterative, and it should be refined and improved throughout your production lifetime. Understanding how efficient your current architecture is in relation to your goals can remove unneeded expense.

Adopt the right consumption model for your use case. You might want to adopt a model where you pay only for the resources you use. Consider using managed services because they operate at cloud scale, and they can offer a lower cost per transaction or service.

For more information, see the Cost Optimization pillar on the content resources page of your online course.



The Sustainability pillar addresses the ability to build architectures that maximize efficiency and reduce waste. The discipline of sustainability addresses the long-term environmental, economic, and societal impact of your business activities. You should understand the impact of your workloads and work to reduce the downstream impact.

Sustainability in the cloud is a continuous effort. It focuses primarily on energy reduction and efficiency across all components of a workload by achieving the maximum benefit from the resources provisioned and minimizing the total resources required. This effort can include the initial selection of an efficient programming language, adoption of modern algorithms, and use of efficient data storage techniques. It can also include deploying to correctly sized and efficient compute infrastructure and minimizing requirements for high-powered end-user hardware.

For more information, see the Sustainability pillar on the content resources page of your online course.

Using the AWS WA Tool



AWS Well-Architected Tool

- Helps you review the state of your workloads and compares them to the latest AWS architectural best practices
- Gives you access to knowledge and best practices used by AWS architects when you need it
- Delivers an action plan with step-by-step guidance on how to build better workloads for the cloud
- Provides a consistent process for you to review and measure your cloud architectures

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If you want help designing a well-architected solution, you can use the AWS Well-Architected Tool. The AWS WA Tool is a self-service tool that provides you with on-demand access to current AWS best practices. These best practices can help you build secure, high-performing, resilient, and efficient application infrastructure on AWS.

The AWS WA Tool helps you review the state of your workloads and compare them to the latest AWS architectural best practices. It gives you access to knowledge and best practices used by AWS architects when you need it.

The AWS WA Tool is available in the AWS Management Console. You define your workload and answer a series of questions in the areas of operational excellence, security, reliability, performance efficiency, and cost optimization. The AWS WA Tool then delivers an action plan with step-by-step guidance on how to improve your workload for the cloud.

The AWS WA Tool provides a consistent process for you to review and measure your cloud architectures. With the tool, you can gather data and get recommendations to do the following:

- Minimize system failures and operational costs.
- Dive deep into business and infrastructure processes.
- Provide best practice guidance.
- Deliver on the cloud computing value proposition.

You can use the results the tool provides to identify next steps for improvement, drive architectural decisions, and bring architecture considerations into your corporate governance process.

For more information, see the *AWS Well-Architected Tool User Guide* on AWS Documentation.

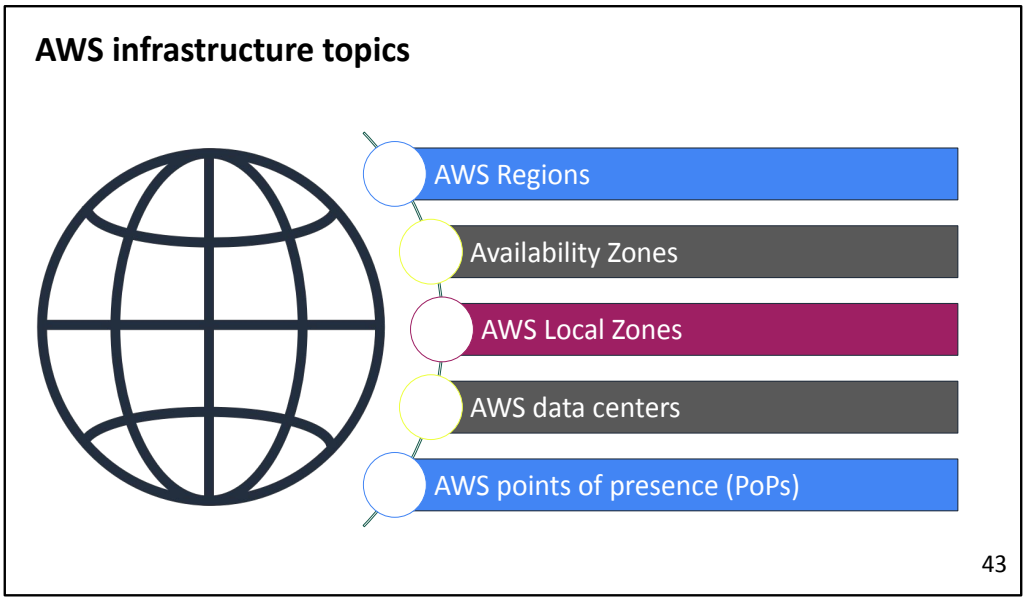
Key takeaways: AWS Well-Architected Framework

- The AWS Well-Architected Framework provides a consistent approach to evaluate cloud architectures and guidance to help implement designs.
- The AWS Well-Architected Framework is organized into six pillars.
- Each pillar documents a set of foundational questions you can use to understand if a specific architecture aligns well with cloud best practices.
- The AWS WA Tool helps you review the state of your workloads and compares them to the latest AWS architectural best practices.

AWS Global Infrastructure

Introducing Cloud Architecting

This section introduces the AWS Global Infrastructure.



See AWS Global Infrastructure on page: https://aws.amazon.com/about-aws/global-infrastructure

The AWS Global Cloud Infrastructure is a secure, extensive, and reliable cloud platform, offering more than 200 fully featured services from data centers globally. The AWS Cloud infrastructure spans 117 Availability Zones in 37 geographic regions around the world. This gives you the ability to launch highly available, scalable, and flexible architectures near your customers for lower latency and increased performance.

As you design and build architectures, you need to consider which region to deploy them. An AWS Region is a physical geographical location with two or more Availability Zones. Availability Zones, in turn, consist of one or more data centers. Your decisions should be based on the business requirements and the needs of the workload.

For example, if low latency is a requirement for your workload, you should consider using AWS points of presence (PoPs) or regional edge caches. The AWS PoP network sits at the edge of the network to reduce latency. Using these edge locations keeps your popular data close to your customers.

In this section, you learn about the AWS infrastructure, including Regions, availability zones, AWS Local Zones, AWS data centers, and AWS PoPs.

Selecting Regions

- A Region is a geographical area.
- Each Region usually consists of two or more Availability Zones.
- Communication between Regions uses AWS backbone network infrastructure.
- You enable and control data replication across Regions.



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A Region is a physical geographical location with two or more Availability Zones. Availability Zones, in turn, consist of one or more data centers.

Regions are connected to multiple internet service providers (ISPs). Regions are also connected to a private global network backbone, which provides lower cost and more consistent cross-Region network latency when compared with the public internet.

Regions that were introduced before March 20, 2019, are enabled by default. Regions that were introduced after March 20, 2019—such as Asia Pacific (Hong Kong) and Middle East (Bahrain)—are disabled by default. You must enable these Regions before you can use them. You can use the AWS Management Console to enable or disable a Region.

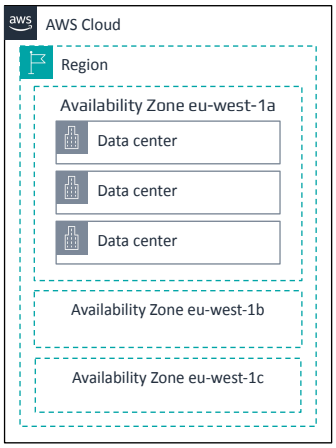
Some Regions have restricted access. The isolated AWS GovCloud (US) Regions are designed to make it possible for US government agencies and customers to move sensitive workloads into the cloud by addressing their specific regulatory and compliance requirements.

To achieve fault tolerance and stability, Regions are isolated from one another. Resources in one Region are not automatically replicated to other Regions. When you store data in a specific Region, it is not replicated outside that Region. It is your responsibility to replicate data across Regions if your business needs require it. AWS provides information about the country and—where applicable—the state where each Region resides. You are responsible for selecting the Region where you should store data based on your compliance and network latency requirements.

AWS products and services are available by Region, so you might not have access to all Regions available for a given service.

Selecting Availability Zones

- Each Availability Zone includes the following:
  - It is made up of one or more data centers.
  - It is designed for fault isolation.
  - It is interconnected with other Availability Zones in a Region using high-speed private links.
- For certain services, you can choose your Availability Zones.
- AWS recommends replicating across Availability Zones for resiliency.



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Each Region consists of two or more isolated locations that called Availability Zones. Each Availability Zone comprises one or more data centers, with some Availability Zones having as many as six data centers. However, no data center can be a part of two Availability Zones.

Each Availability Zone is designed as an independent failure zone. This means that Availability Zones are physically separated in a typical metropolitan Region. They are also located in lower-risk floodplains (specific flood-zone categorization varies by Region). Availability Zones have a discrete, uninterruptible power supply and on-site backup generation facilities. In addition, they are each fed by different grids from independent utilities to further reduce single points of failure. Availability Zones are all redundantly connected to multiple tier-1 transit providers.

An Availability Zone is the most granular level of specification that you can make for certain services, such as Amazon EC2.

You are responsible for selecting the Availability Zones where your systems will reside. Systems can span multiple Availability Zones. You should design your systems to survive temporary or prolonged failure of an Availability Zone if a disaster occurs. Distributing applications across multiple Availability Zones means they can remain resilient in most failure situations, including natural disasters or system failures.

Using Local Zones

- Local Zones make it possible for you to run latency-sensitive portions of applications closer to end users and resources in a specific geography.
- They are an extension of a Region.
- With Local Zones, you can place AWS compute, storage, database, and other select services closer to large population, industry, and IT centers where no Regions exist today.
- Local Zones are managed and supported by AWS.

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Local Zones are a type of AWS infrastructure deployment. They place AWS compute, storage, database, and other select services closer to large population, industry, and IT centers where no Regions exist today. With Local Zones, you can run latency-sensitive portions of applications closer to end users and resources in a specific geography. You can use Local Zones to deliver single-digit millisecond latency for use cases such as media and entertainment content creation, real-time gaming, reservoir simulations, electronic design automation, and machine learning (ML).

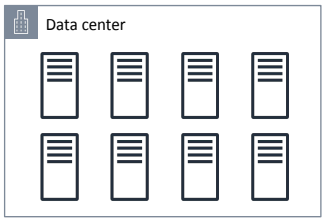
Each Local Zone location is an extension of a Region. You can run latency-sensitive applications in a Local Zone by using AWS services such as Amazon EC2, Amazon Virtual Private Cloud (Amazon VPC), Amazon Elastic Block Store (Amazon EBS), Amazon FSx, and ELB in geographic proximity to end users. Local Zones provide a high-bandwidth, secure connection between local workloads and workloads that run in the Region. With Local Zones you can seamlessly connect back to your other workloads that are running in AWS. You can connect to the full range of in-Region services through the same APIs and toolsets.

Local Zones are managed and supported by AWS, and they provide you with all the elasticity, scalability, and security benefits of the cloud. With Local Zones, you can build and deploy latency-sensitive applications closer to your end users by using a consistent set of AWS services. You can also scale up or scale down, and you pay only for the resources you use.

For more information, see AWS Local Zones on the content resources page of your online course.

Role of AWS data centers

- Data centers are where the data resides and data processing occurs.
- A data center typically has tens of thousands of servers.
- All data centers are online and serving customers.
- AWS custom network equipment includes the following:
  - Is sourced from multiple ODMs
  - Has a customized network protocol stack

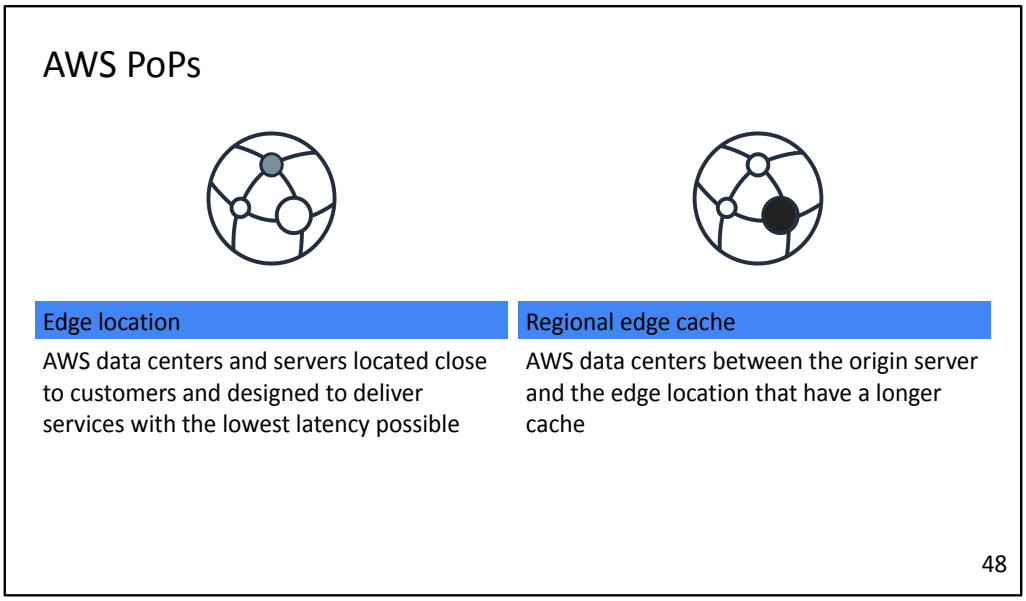


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The foundation for the AWS infrastructure is the data centers. You do not specify a data center for the deployment of resources. However, a data center is the location where the actual data resides. Amazon operates state-of-the-art, highly available data centers. Though rare, failures that affect the availability of instances in the same location can occur. If you host all your instances in a single location that is affected by such a failure, none of your instances will be available.

All data centers are online and serving customers. In case of failure, automated processes move customer data traffic away from the affected area. Core applications are deployed in an N+1 configuration. In the event of a data center failure, there is sufficient capacity for traffic to be load balanced to the remaining sites.

AWS uses custom network equipment sourced from multiple original device manufacturers (ODMs). ODMs design and manufacture products based on specifications from another company. The other company then rebrands the products for sale.



To deliver content to end users with lower latency, CloudFront uses a global network that includes more than 410 PoPs. The PoPs are comprised of 400 edge locations and 13 regional mid-tier caches. These PoPs are the initial destination for a CloudFront request.

The edge locations make sure that popular content can be served quickly to customers. Regional edge caches bring more of your content closer to customers even if it is not popular enough to stay at an edge location. This helps reduce latency for customers. Regional edge caches increase efficiency and are transparent to the end user.

Edge locations are located in North America, Europe, Asia, Australia, South America, the Middle East, Africa, and China. Edge locations support AWS services like Amazon Route 53, AWS Global Accelerator, and CloudFront.

Regional edge caches are used by default with CloudFront. They are used when you have content that is not accessed frequently enough to remain in an edge location. Regional edge caches absorb this content and provide an alternative to fetching the content from the origin server.

For more information, see *Points of Presence* on AWS website.

Key takeaways: AWS Global Infrastructure

- The AWS Global Infrastructure consists of Regions, Availability Zones, and edge locations.
- Your choice of a Region is typically based on compliance requirements or to reduce latency.
- Each Availability Zone is physically separate from other Availability Zones and has redundant power, networking, and connectivity.
- Edge locations and regional edge caches improve performance by caching content closer to users.